

Question				Marking details	Marks available																	
					AO1	AO2	AO3	Total	Maths	Prac												
5	(a)	(i)		D <u>B</u> E A C			1	1		1												
		(ii)		magnesium	1			1														
		(iii)		<table><tr><td>volume of soap solution</td><td>✓</td></tr><tr><td>type of water</td><td></td></tr><tr><td>type of soap solution</td><td>✓</td></tr><tr><td>volume of water</td><td>✓</td></tr><tr><td>height of lather</td><td></td></tr><tr><td>width of test tube</td><td>✓</td></tr></table> award (2) for all 4 correct with no other ticks award (1) for any 2 or 3 deduct (1) per incorrect box ticked	volume of soap solution	✓	type of water		type of soap solution	✓	volume of water	✓	height of lather		width of test tube	✓			2	2		2
volume of soap solution	✓																					
type of water																						
type of soap solution	✓																					
volume of water	✓																					
height of lather																						
width of test tube	✓																					
	(b)	(i)		54°C no tolerance		1		1	1													
		(ii)		solubility of both increases as the temperature increases (1) award (1) for any of following (solubility of) copper(II) sulfate increases much more (than that of sodium chloride) (solubility of) sodium chloride increases much less (than that of copper(II) sulfate) (solubility of) copper(II) sulfate increases a lot and that of sodium chloride increases only slightly comparison needed		2		2														

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		290 g (2) if incorrect award (1) for either of following 29 g (correct reading from graph) any value multiplied by 10		2		2	2	
	(c)	(i)		1		1		1		
		(ii)		174 (2) if incorrect award (1) for either of following $(39 \times 2) + 32 + (4 \times 16)$ $2\text{ K} + 1\text{ S} + 4\text{ O}$		2		2	2	
				Question 5 total	1	8	3	12	5	3